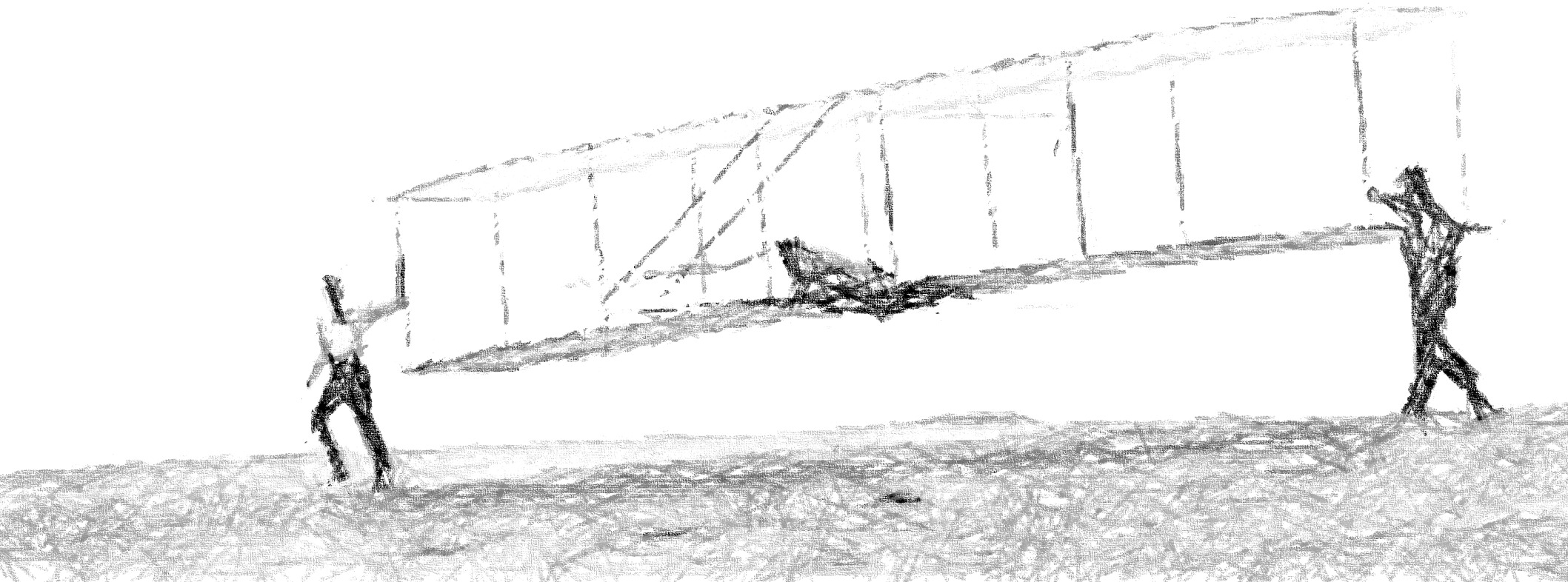




Mechanics applied to Aerospace Engineering

Course 251-14165 – BSc Aerospace Engineering

About the course



Objectives of the course

- Master the key concepts and tools of the mechanics of point particles and rigid bodies
- Acquire the abilities to pose and solve classical mechanics problems:
 - Choose adequate reference systems to study the motion
 - Identify the number of degrees of freedom of a problem and choose adequate coordinates to study it
 - Study the kinematics of the problem (i.e., express velocities and accelerations in terms of the chosen coordinates)
 - Identify, classify, and write the forces acting on the problem
 - Pose the equations of motion of the problem and solve them
 - Identify any conservation laws that may exist (e.g. angular momentum, mechanical energy)
- Learn about fundamental applications of Newtonian mechanics to aerospace engineering



What you should know already

- Expected previous knowledge (important!):
 - Mathematics and physics knowledge, in particular:
 - Linear algebra (bases, rotations, orthogonality, vector projections, matrices, eigenvalues)
 - Vector calculus (differentiation, integration, differential equations)
 - Analytic geometry (curves, surfaces, trigonometry...)

You should not just “be familiar” with these concepts. They should be second-nature to you. Ask us for help or references if you feel at any point that you have troubles with these or other topics!

“Your success in this course depends 95% on your mastery of these subjects”

About the course

- Course has 6 ECTS:
 - 28 lectures + 4 labs (90 min each) + Home work time by the student
 - Expected ~150 h total workload on the student.

Individual and group student work is essential for the correct completion of the course. Roughly, 2/3 of the work is carried out outside of lectures

- All course-related information and communication with the students will be uploaded to the Aula Global page. Please check it often!
 - Lecture notes, videos, problems and exercises, lab guidelines
 - Updated course schedule, important messages, and any lecture changes
 - Weekly non-evaluative additional practice exercises
 - Grades of quizzes and exams

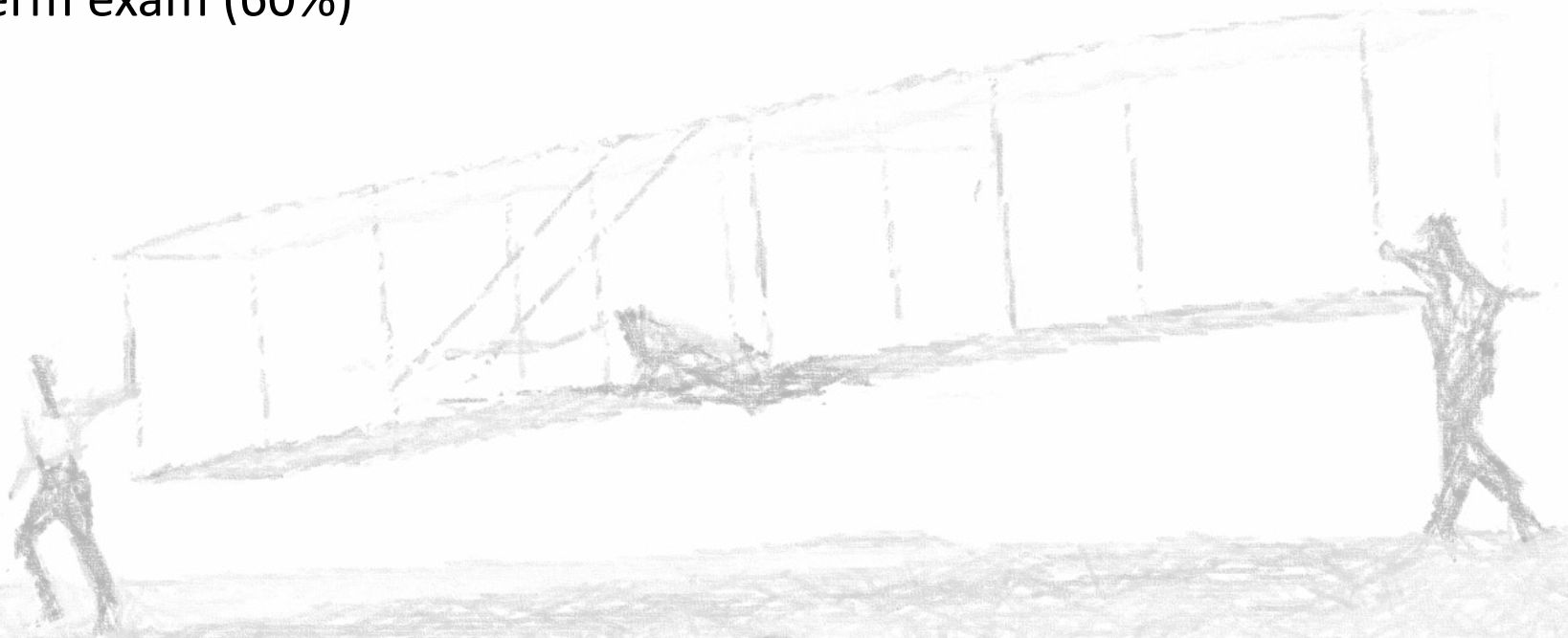
Lab sessions

The course comprises a total of 4 lab sessions

- Lab sessions will be carried out in groups of 3 students
 - Lab groups will be assigned soon (see Aula Global)
- Each group will complete and submit a report for each lab session after one week
 - Good scientific reporting practices are mandatory. A document containing report guidelines can be found in Aula Global
 - Help with Matlab function ode45 will also be available in Aula Global
- Students retaking the course who have a >5 grade in their lab sessions in previous years do not have to redo them again
 - If you do not want to redo them and would like to keep your lab grade, email Alberto (almarinc@pa.uc3m.es) before end of week 1
 - Failure to do so means you will need to repeat the lab sessions
- The use of Generative AI is not allowed in this course; any usage in the lab reports will be penalized academically.
(You should focus on *learning*, at this stage!)

Grading system

- In order to pass the course, two requirements need to be met:
 1. To have a minimum grade of 4.0/10 in each of the three parts (theory, problems) of the end-of-term exam;
 2. To have a minimum overall grade of 5.0/10 (weighing 60% the end-of-term exam and 40% the continuous evaluation).
- Continuous evaluation (40%):
 - 20%: Lab/computer sessions + reports
 - 20%: quizzes (2 in total)
- End-of-term exam (60%)



Course materials and bibliography

- Lecture notes are provided with a summary of course contents
 - Lecture notes do not replace books. Reading the bibliography is necessary to fully understand the course; i.e., book contents can be part of the final exam
- A collection of exercises and problems to be solved in class
 - More practice exercises and problems can be found in the literature
- Optional materials: Additional videos with problem solving are also available

Basic bibliography

- J. H. Ginsberg, Engineering Dynamics, Cambridge Univ. Press 2007

Additional bibliography

- H. Goldstein, Classical Mechanics, Higher Education Press 2002
- M. Prieto, Mecánica Racional I y II (in Spanish only), ADI 1986
- A.C. Kermode, Mechanics of Flight, 12th Ed., Pearson 2012
- W.T. Thomson, Introduction to Space Dynamics, Dover 1986

Other, online resources to get up to speed in math:

- Linear algebra and calculus tutorials: [3blue1brown linear algebra](#), [calculus](#). Worth a look!

Session plan

- Updated session plan will be posted on AulaGlobal . **Please check session changes**

Information for students



URL
Course description and time plan



URL
Recommended bibliography



URL
Subject guide in Engineering



FILE
Weekly planning (last updated: 2024-09-06)

Course Instructors and office hours

Please address the corresponding professor for office hours by sending an email; we will answer you to set an appointment:

- Alberto Marín (course coordinator)
almarinc@pa.uc3m.es
Office hours: theory, problems, labs 3 and 4
- Borja Bayón
bbayon@pa.uc3m.es
Office hours: problems
- Iñaki Fernández
inaferna@ing.uc3m.es
Office hours: lab 1
- Hugo Bergerioux
hbergeri@ing.uc3m.es
Office hours: lab 2

Office Hours and Student Feedback System

- Do use the office hours to solve your doubts and questions!
 - Simply send us an email and we will arrange a meeting
- We need your feedback (good or bad) to improve this course.
 - When you detect something that could be improved, please talk directly with us or send us an email (almarinc@pa.uc3m.es)

Juan Benet II

